Energy-Balanced Opportunistic Routing in Drifting Underwater Wireless Sensor Networks via Multi-Objective SARSA

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***Abstract*—Underwater Wireless Sensor Networks (UWSNs) are used for oceanographic monitoring and a range of other underwater applications. Their communication environment, however, is constrained by limited bandwidth, long propagation delay, restricted energy availability, intermittent connectivity, and node movement caused by ocean currents. These factors make it difficult to maintain reliable packet forwarding while distributing energy use across the network. Reinforcement-learning-based protocols such as QELAR and EDORQ have improved forwarding decisions, but their use of off-policy Q-learning can produce optimistic action-value estimates when the topology changes.**

This paper presents an Energy-Balanced Multi-Objective SARSA-based Opportunistic Routing Protocol for drifting UWSNs. The method uses on-policy SARSA together with a reward function that accounts for depth progress, residual energy, local variation in neighboring-node energy, and propagation distance. An arctangent-based holding-time rule is used to coordinate opportunistic forwarding and limit duplicate transmissions. A void-recovery procedure is also included for situations in which no suitable shallower forwarder is available.

The protocol is evaluated in a $(500 \times 500 \times 500)$ $\mathbf{m}^3$ drifting underwater environment using a 10-kbps acoustic data rate, a 100-m transmission range, and a propagation speed of 1500 m/s. At 200 nodes, the proposed method obtains a packet delivery ratio (PDR) of 78.4%, compared with 72.1% for EDORQ and 54.0% for DBR. At 800 nodes, the corresponding PDR values are 95.6%, 92.1%, and 86.3%. The results show higher delivery rates for the proposed protocol in the tested node-density settings, while the reward also accounts for how energy is distributed across neighboring nodes.

***Index Terms*—Underwater Wireless Sensor Networks, underwater acoustic communication, opportunistic routing, reinforcement learning, SARSA, Q-learning, energy balancing, routing void, node mobility.**

I. Introduction

Underwater Wireless Sensor Networks (UWSNs) are useful for oceanographic data collection, environmental monitoring, and underwater disaster prevention. Communication in these networks differs considerably from communication in terrestrial wireless networks. The available bandwidth is limited, propagation delay is relatively high, and sensor nodes can change position as they are carried by ocean currents. These characteristics place strong constraints on the design of routing protocols [1]. Depth-Based Routing (DBR) is one of the commonly used localization-free approaches because forwarding decisions can be made from depth information obtained through pressure sensors rather than complete geographical coordinates [2]. Opportunistic routing (OR) can further improve delivery by allowing more than one neighboring node to act as a potential forwarder over the broadcast acoustic channel [3].

Reinforcement learning (RL) has also been applied to opportunistic routing to make forwarding decisions using information gathered during network operation. Q-learning-based approaches can adapt their decisions from observed rewards, but the off-policy update uses the maximum estimated value at the next state. When the topology changes because of node drift, an action with a high estimated value may not always be the most suitable action under the current forwarding policy. This can contribute to unstable forwarding choices and can make routing voids more difficult to handle. At the same time, repeated forwarding through a small set of favorable nodes can concentrate energy consumption in particular parts of the network.

The approach studied here uses SARSA (State-Action-Reward-State-Action), an on-policy reinforcement learning method. Its update uses the next action chosen by the current policy rather than the largest estimated value over all possible actions. As a result, the learning update follows the forwarding decisions that are actually made, which is useful when node positions change with the drift.

The reward used by the protocol brings four quantities into the forwarding decision: progress toward the surface, residual energy, variation in the energy of neighboring nodes, and propagation distance. Including the variance term makes the decision sensitive to local energy imbalance instead of considering only the selected node. The protocol also maps the learned SARSA value to an arctangent-scaled holding time. A candidate with a more favorable value waits less, and the remaining candidates can cancel their transmissions after hearing the forwarding event.

II. Literature Survey

Routing in UWSNs has progressed from simple forwarding schemes toward approaches that make greater use of network

conditions and adaptive decision making. Vector-Based Forwarding (VBF) restricts forwarding to nodes located within a virtual routing pipe between the source and destination. This can reduce unnecessary participation, but its performance depends on node density and the choice of pipe width [6]. DBR instead uses depth information to identify nodes that can make progress toward the water surface [2]. Although this avoids the need for full localization, repeated use of nodes nearer to the surface can result in redundant transmissions and uneven energy consumption.

Reinforcement learning has been introduced to address some of these limitations. QELAR uses Q-learning to incorporate energy information into routing decisions with the aim of extending network lifetime [7]. Other Q-learning-based methods have considered delay, node mobility, or cooperative forwarding. The literature also shows that exchanging learning information can introduce additional overhead, while approaches that depend on accurate position estimates can be difficult to apply in underwater environments [8]. QMCR, for example, applies Q-learning to multi-hop cooperative routing, but its forwarding strategy still gives substantial importance to distance-related choices rather than explicitly measuring local variation in residual energy [9].

Void handling is another important aspect of underwater routing. A routing void occurs when a forwarding node cannot find a suitable neighbor that provides the required progress toward the sink. Existing approaches use different recovery strategies, including pressure information, alternate candidate selection, and topology-aware forwarding. EDORQ combines depth-based opportunistic routing with Q-learning, residual energy, void detection, and Q-value-based holding time. Its reward design and recovery mechanism are particularly relevant to the forwarding framework considered in this work [12].

The proposed method differs from these approaches mainly in the learning update and the treatment of local energy distribution. It replaces the off-policy Q-learning update with on-policy SARSA and includes neighbor-energy variance directly in the reward. The intention is to consider routing progress and node energy together rather than repeatedly favoring the same locally attractive relays.

III. Problem Statement

UWSNs operate with limited bandwidth, relatively long acoustic propagation delay, constrained battery energy, and node movement caused by ocean currents. Because the topology is not static, a forwarding choice that is suitable at one moment may become less effective later. Conventional routing can therefore encounter routing voids, repeated use of particular relays, and uneven energy depletion. A useful forwarding mechanism should consequently consider not only progress toward the sink but also the remaining energy of candidate nodes, the distribution of energy among their neighbors, and communication delay.

Q-learning-based routing introduces another consideration. Its off-policy update evaluates the next state using the maximum estimated action value. In a changing underwater topology, this estimate can favor an action that is not necessarily the one followed by the current policy. In addition, selecting the same energy-rich or geometrically favorable nodes repeatedly can concentrate traffic and create energy holes, especially in regions close to surface sinks. The routing problem addressed in this work is therefore to balance forwarding reliability and progress with energy distribution while remaining capable of responding to temporary topology changes.

A. System and Acoustic Energy Model

1) Network and Drift Model

We consider a three-dimensional underwater region with volume

$$V = 500 \times 500 \times 500\,\mathrm{m}^3. \tag{1}$$

Sensor nodes $(\mathcal{N})$ are randomly deployed in the underwater region, while multiple sinks are placed at the water surface. Ocean currents cause the sensor nodes to drift horizontally, represented by a current velocity vector $(\vec{v}_c)$, so their positions can change during operation. Each node has a maximum communication range of $(R_{\max} = 100\,\mathrm{m})$, consistent with the LinkQuest UWM1000 modem parameters used as a reference [4].

2) Acoustic Channel and Energy Model

The attenuation of an acoustic signal over distance (d) at frequency (f) is represented using Thorp's absorption model:

$$10 \log A(d, f) = k \cdot 10 \log d + d \cdot \alpha(f), \tag{2}$$

where (k) denotes the spreading factor and is taken as (1.5) for practical spreading, while $(\alpha(f))$ is the absorption coefficient in $(\mathrm{dB/km})$. The distance used for the absorption component is expressed in kilometres.

Energy-aware and opportunistic routing has received considerable attention in UWSNs. QELAR uses residual-energy information in its routing decision, while EDORQ combines Q-learning with depth-based opportunistic forwarding and void handling. Other studies have addressed energy imbalance, node mobility, and routing recovery. These works motivate the use of multiple routing objectives in the proposed method, which considers depth progress, residual energy, local neighbor-energy variance, and propagation distance within a single reward.

The energy required to transmit a packet of size (L) is

$$E_{\mathrm{tx}} = P_{\mathrm{tx}} \left(\frac{L}{\mathrm{BitRate}} \right), \tag{3}$$

where (P_{tx}) is the transmission power, (L) is the packet length in bits, and $(\mathrm{BitRate})$ is the acoustic data rate.

Similarly, the energy consumed while receiving the packet is

$$E_{\mathrm{rx}} = P_{\mathrm{rx}} \left(\frac{L}{\mathrm{BitRate}} \right), \tag{4}$$

where (P_{rx}) is the receiving power.

B. Proposed Multi-Objective SARSA Protocol

1) SARSA Update Formulation

The proposed protocol uses SARSA in place of the off-policy Q-learning update

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right]. \quad (5)$$

In SARSA, the sensor node acts as the learning agent and updates its value using the current state (s), selected action (a), received reward (r), next state (s'), and next action (a'):

$$Q(s_i,a_j) \leftarrow Q(s_i,a_j) + \alpha \left[R(s_i,a_j) + \gamma Q(s_j,a_k) - Q(s_i,a_j) \right]. \quad (6)$$

Here, (s_i) denotes the current routing state, (a_j) the selected forwarding action, $(R(s_i,a_j))$ the immediate reward, (s_j) the next state, and (a_k) the next action chosen by the current policy. The parameters (α) and (γ) are the learning rate and discount factor, respectively.

The next action is selected using an (ϵ)-greedy policy. With probability $(1-\epsilon)$ the candidate with the largest estimated value is selected, while with probability (ϵ) an admissible candidate is chosen randomly. Since the SARSA target uses the action actually selected by the policy, the update follows the experienced routing trajectory rather than replacing the next action with the maximum possible value.

2) State and Action Representation

The routing problem is represented as a discrete Markov decision process (MDP):

$$\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R} \rangle, \quad (7)$$

where $(\mathcal{S})$ is the set of routing states, $(\mathcal{A})$ is the set of forwarding actions, $(\mathcal{P})$ represents state transitions, and $(\mathcal{R})$ is the reward function. A state $(s_i \in \mathcal{S})$ represents the condition in which a packet has reached node (n_i).

For node (n_i), the admissible forwarding actions are defined by

$$\mathcal{A}(s_i) = \{ a_j \mid n_j \in \mathcal{N}_i \}, \quad (8)$$

where $(\mathcal{N}_i)$ is the one-hop neighborhood and (a_j) represents choosing node (n_j) as the next forwarder.

Candidate selection primarily favors positive progress toward the surface. For nodes (n_i) and (n_j), the depth advancement is

$$\Delta d_{ij} = d_i - d_j, \quad (9)$$

where (d_i) and (d_j) denote their respective depths. A positive value indicates that the candidate is closer to the water surface.

3) Multi-Objective Reward Function

The proposed reward combines forwarding progress, residual energy, local energy balance, and propagation distance:

$$R(s_i,a_j) = w_1 \left(\frac{\Delta d}{R_{\max}} \right) + w_2 \left(\frac{E_{\text{res}}}{E_{\text{init}}} \right) - w_3 \left(\frac{\sigma^2(E_{\text{neighbors}})}{E_{\text{init}}^2} \right) - w_4 \left(\frac{\text{dist}}{v_s T_{\max}} \right). \quad (10)$$

Here, $(w_1, w_2, w_3, w_4 \geq 0)$ determine the relative importance of the four objectives. The first term rewards progress toward the surface, the second favors candidates with greater residual energy, the third penalizes uneven energy among neighboring nodes, and the fourth accounts for propagation distance.

The distance between the current node and candidate is

$$\text{dist} = |\mathbf{p}_i - \mathbf{p}_j|_2, \quad (11)$$

where $(\mathbf{p}_i)$ and $(\mathbf{p}_j)$ are their instantaneous positions. The acoustic propagation velocity is $(v_s = 1500\,\text{m/s})$, and $(T_{\max})$ is the maximum forwarding coordination interval.

The local variance term is included to discourage repeated forwarding through a small group of nodes. In this way, the reward does not depend only on the energy of the individual candidate but also reflects how evenly energy is distributed in its neighborhood.

4) Timer-Based Forwarding Coordination

After candidate values have been obtained, each candidate is assigned a holding time based on its learned value:

$$H_j = \left[1 - \frac{2}{\pi} \arctan\left(Q(s_i,a_j) \right) \right] T_{\max}, \quad (12)$$

where (H_j) is the waiting time of candidate (n_j). The arctangent mapping limits the effect of very large positive or negative Q-values. A candidate with a larger learned value therefore receives a shorter holding time and gets the first opportunity to forward the packet. Other candidates can overhear that transmission and suppress their own forwarding attempts.

The maximum coordination interval is selected as

$$T_{\max} = \frac{2R_{\max}}{v_s}, \quad (13)$$

where $(R_{\max})$ is the maximum acoustic transmission range and (v_s) is the propagation velocity.

C. Routing Procedure

The proposed routing process can be summarized as follows:

1) A source node creates a data packet and broadcasts it to its one-hop neighbors.
2) Each receiving node obtains the relevant depth information from the packet and its pressure sensor.
3) Nodes satisfying the forwarding condition are added to the candidate set.
4) Each candidate obtains its current state-action value $(Q(s_i,a_j))$.
5) The candidate evaluates the multi-objective reward using depth progress, residual energy, local energy variance, and propagation distance.
6) An action is selected according to the (ϵ)-greedy policy.

7) The next state and selected next action are used in the SARSA update.

$$Q(s_i, a_j) \leftarrow Q(s_i, a_j) + \alpha \left[R(s_i, a_j) + \gamma Q(s_j, a_k) - Q(s_i, a_j) \right]. \tag{14}$$

8) The candidate calculates its arctangent-scaled holding time.
9) The candidate whose timer expires first forwards the packet.
10) Other candidates suppress their transmissions after overhearing the forwarding event.
11) If no successful forwarding is overheard within $(T_{\max})$, void recovery is activated.
12) The process continues until a surface sink receives the packet or the forwarding limit is reached.

D. Performance Metrics

1) Packet Delivery Ratio

Packet delivery ratio (PDR) is defined as

$$\mathrm{PDR} = \frac{N_{\mathrm{delivered}}}{N_{\mathrm{generated}}}, \tag{15}$$

where $(N_{\mathrm{delivered}})$ is the number of packets received by surface sinks and $(N_{\mathrm{generated}})$ is the number generated by source nodes.

2) Total Energy Consumption

Total energy consumption (TEC) is the sum of transmission, reception, and idle energy used by all sensor nodes:

$$\mathrm{TEC} = \sum_{i=1}^{N} \left(E_{\mathrm{tx},i} + E_{\mathrm{rx},i} + E_{\mathrm{idle},i} \right). \tag{16}$$

3) Energy Balance Index

Energy balance is measured with Jain's fairness index applied to residual energy:

$$\mathrm{EBI} = \frac{\left(\sum_{i=1}^{N} E_i \right)^2}{N \sum_{i=1}^{N} E_i^2}. \tag{17}$$

The index lies in $(0, 1]$. A value closer to one indicates a more even distribution of residual energy among the nodes.

4) Average End-to-End Delay

Average packet delay is defined as

$$\mathrm{APD} = \frac{1}{N_{\mathrm{delivered}}} \sum_{p=1}^{N_{\mathrm{delivered}}} \left(t_p^{\mathrm{sink}} - t_p^{\mathrm{source}} \right), \tag{18}$$

where (t_p^{source}) is the generation time of packet (p) and (t_p^{sink}) is its reception time at a surface sink.

5) Average Network Overhead

Average network overhead is defined as

$$\mathrm{ANO} = \frac{N_{\mathrm{forwarding}}}{N_{\mathrm{delivered}}}, \tag{19}$$

where $(N_{\mathrm{forwarding}})$ denotes the total forwarding transmissions, including data and control transmissions, and $(N_{\mathrm{delivered}})$ is the number of successfully delivered packets.

IV. Proposed Methodology

The proposed Multi-Objective SARSA protocol is intended for drifting UWSNs in which node movement, acoustic delay, energy imbalance, and routing voids can affect packet delivery. The methodology combines on-policy SARSA learning with energy-aware candidate selection, opportunistic forwarding, and a recovery procedure for local voids.

A. Network and Channel Model

A three-dimensional underwater network is considered within a $(500 \times 500 \times 500)\,\mathrm{m}^3$ region. Sensor nodes drift horizontally with the modeled ocean current, while multiple sinks on the surface collect the sensed information. Acoustic communication uses a maximum range of 100 m and a propagation speed of 1500 m/s. Transmission, reception, and idle-state energy are included in the energy model.

B. State and Action Representation

Each node considers its one-hop neighbors as possible forwarding candidates. The routing state reflects depth, residual energy, and neighborhood conditions. The depth progress from node i to candidate j is

$$\Delta d_{ij} = d_i - d_j, \tag{20}$$

where positive (Δd_{ij}) represents progress toward the surface sink.

C. Multi-Objective SARSA

Rather than using the maximum next-state Q-value as in Q-learning, the proposed protocol updates the value using the action actually selected under the current policy:

$$Q(s_i, a_j) \leftarrow Q(s_i, a_j) + \alpha \left[R(s_i, a_j) + \gamma Q(s_j, a_k) - Q(s_i, a_j) \right]. \tag{21}$$

An (ϵ)-greedy policy provides a balance between exploration and exploitation.

D. Multi-Objective Reward Function

The reward combines forwarding progress, residual energy, neighborhood energy balance, and propagation delay:

$$\begin{aligned} R(s_i, a_j) = {} & w_1 \frac{\Delta d}{R_{\max}} + w_2 \frac{E_{\mathrm{res}}}{E_{\mathrm{init}}} \\ & - w_3 \frac{\sigma^2(E_{\mathrm{neighbors}})}{E_{\mathrm{init}}^2} - w_4 \frac{\mathrm{dist}}{v_s T_{\max}}. \end{aligned} \tag{22}$$

This formulation favors candidates that provide useful forwarding progress and have sufficient energy, while discouraging highly uneven local energy distributions and longer propagation distances.

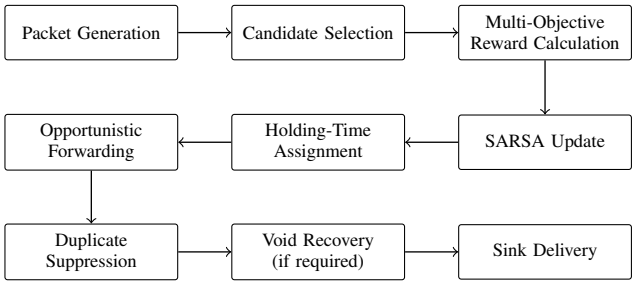


Fig. 1. Routing workflow

E. Void Recovery and Holding-Time Mechanism

When no suitable forwarder can provide the required progress, the protocol enters void recovery and permits an alternative neighboring node to be considered. For opportunistic forwarding, the candidates use the following arctangent-based holding time:

$$H_j = \left[1 - \frac{2}{\pi} \arctan\left(Q(s_i, a_j)\right)\right] T_{\max}. \tag{23}$$

Candidates with larger learned values wait for less time and are therefore more likely to forward first. When another candidate successfully forwards the packet, the remaining candidates suppress duplicate transmissions.

V. Results and Analysis

Monte Carlo simulations were used to examine the behavior of the routing protocols as the number of sensor nodes increased from 200 to 800. With a larger node population, more neighboring relays are likely to fall within the 100-m acoustic range, which gives opportunistic forwarding a larger set of candidates.

At 200 nodes, DBR records a PDR of 54.0%, while EDORQ reaches 72.1%. The proposed Multi-Objective SARSA protocol records 78.4%, which is 6.3 percentage points above EDORQ and 24.4 percentage points above DBR. At 800 nodes, the PDR values increase to 95.6% for the proposed method, 92.1% for EDORQ, and 86.3% for DBR.

Two parts of the design help explain this trend. The SARSA update follows the next action selected by the current policy instead of taking the maximum estimated value at that state. In addition, the neighborhood-energy variance term penalizes decisions that repeatedly favor the same relay region. Opportunistic forwarding and the void-recovery step work alongside these terms when the available candidates change with node density.

VI. Discussion and Comparison with Existing Research

A. Comparison with Existing Research

Existing UWSN routing protocols differ in the information they use to select and coordinate forwarding nodes. VBF relies on a vector-based forwarding region, DBR uses depth information, and QELAR and EDORQ incorporate reinforcement learning. The proposed Multi-Objective SARSA approach combines on-policy learning with forwarding progress, residual energy, local energy variance, and propagation distance.

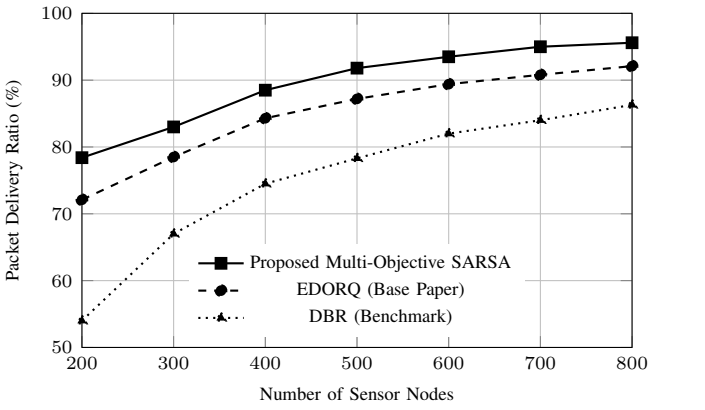


Fig. 2. Packet Delivery Ratio (PDR) versus sensor node density

TABLE I
Comparison of Existing UWSN Routing Protocols

Protocol	Routing Basis	Learning / Forwarding	Energy / Void Handling
VBF	Vector-based	Opportunistic	Basic void tolerance
DBR	Depth-based	Single-path	Limited energy awareness
QELAR	Energy-aware	Q-learning / Adaptive	Residual-energy based
EDORQ	Depth + energy	Q-learning / Opportunistic	Energy + void detection
QTAR	Topology-aware	Q-learning / Opportunistic	Topology + void awareness
Proposed	**Multi-objective**	**SARSA / Opportunistic**	**Energy variance + void recovery**

B. Performance Comparison

The available evaluation compares the proposed protocol with DBR and EDORQ using PDR. At 200 nodes, the proposed method achieves 78.4%, compared with 72.1% for EDORQ and 54.0% for DBR. At 800 nodes, the corresponding values are 95.6%, 92.1%, and 86.3%. Thus, the proposed approach improves over EDORQ by 6.3 percentage points at 200 nodes and by 3.5 percentage points at 800 nodes.

C. Discussion

The PDR trend is consistent with the way the routing decision is constructed. The reward considers not only forwarding progress but also the candidate's residual energy and the energy variation among its neighbors. Opportunistic forwarding keeps multiple candidates available, and the holding-time rule lets the candidate with the more favorable learned value transmit first. If normal depth-based progress cannot continue, the void-recovery step provides another forwarding option.

The results show that the proposed SARSA-based design maintains higher PDR in the tested node-density range, while its reward explicitly includes local energy distribution. The experiments reported here focus on PDR; the other metrics

defined in the model can be used in a broader evaluation of energy consumption, delay, and overhead.

VII. Future Scope

The proposed Multi-Objective SARSA protocol can be extended in several directions:

- **Realistic Simulation:** Implement the protocol in underwater network simulators such as Aqua-Sim or DESERT Underwater to evaluate it under more detailed channel and mobility models.
- **Real Ocean-Current Data:** Use measured ocean-current traces to represent realistic node movement rather than idealized drift patterns.
- **Dynamic Reward Weights:** Adapt reward weights according to residual energy, node density, and current conditions.
- **Multi-Agent Reinforcement Learning:** Extend SARSA toward cooperative multi-agent learning so neighboring nodes can coordinate forwarding decisions.
- **Advanced Acoustic Channel Modeling:** Include Doppler effects, multipath propagation, packet loss, variable acoustic frequency, and time-varying channel conditions.
- **Adaptive Transmission Power:** Add transmission-power control to reduce energy use and extend network lifetime.
- **Scalability Analysis:** Study larger and denser networks to quantify scalability and computational overhead.
- **Real-World Validation:** Test the routing strategy using underwater acoustic modems in practical underwater environments.
- **Parameter Optimization:** Optimize SARSA parameters and reward weights to study the trade-off among delivery reliability, energy consumption, delay, and network lifetime.

VIII. Conclusion

This work developed an energy-balanced opportunistic routing approach for drifting UWSNs using Multi-Objective SARSA. Compared with the Q-learning update used in EDORQ, the proposed method uses an on-policy SARSA update, so the value update follows the action selected by the current forwarding policy.

The reward brings together depth progress, residual energy, local neighbor-energy variance, and propagation distance. The variance term discourages repeated selection of the same forwarding region, and the arctangent-based holding time orders candidate transmissions while reducing duplicate forwarding. When a suitable candidate cannot be found through normal depth-based progress, the protocol invokes void recovery.

In the reported PDR evaluation, the proposed method achieves 78.4% at 200 nodes and 95.6% at 800 nodes, compared with EDORQ values of 72.1% and 92.1%, respectively, and DBR values of 54.0% and 86.3%. The reported PDR values show the potential of combining on-policy learning with local energy-balance information in a drifting UWSN. A wider evaluation is still needed to measure energy consumption, delay, overhead, convergence, and the individual contribution of the routing components under more realistic underwater conditions.

Future work will therefore focus on realistic simulators, measured ocean-current data, more detailed acoustic-channel effects, adaptive reward weights, and practical underwater validation.

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